EVALUATION

I Choose the correct answer:

1. (d) at high pressure the intermolecular interactions become significant
2. (d) inversely proportional to the square root of its molecular weight
3. (c) $\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$
4. (b) exert no attractive forces on each other
5. (a) 1/3
6. (b) Boyle temperature
7. (c) diffusion
8. (b) near the hydrogen chloride bottle
9. (d) units of pressure and volume
10. (c) 8.3 J mol^{-1} K^{-1}
11. (a) Boyle's Law
12. (c) NH_3
13. (d) I, II and III
14. (c) 0.41 dm^3
15. (c) P
16. (b) 4
17. (c) 1/8
18. (b) 1/T
19. (a) P
20. (b) NH_3
21. (c) mol^{-1} L and L^2 atm mol^{-2}
22. (d) both assertion and reasons are false
23. (c) 3.41 g L^{-1}
24. (c)
25. (d) HI

Keys to multiple choice questions:

5. mass of methane
   = mass of oxygen = a

   number of moles of methane = $\frac{a}{16}$

   number of moles of Oxygen = $\frac{a}{32}$

   mole fraction of Oxygen = $\frac{\frac{a}{32}}{\frac{a}{16}+\frac{a}{32}}$

   $=\frac{\frac{a}{32}}{\frac{3a}{32}}$

   $=\frac{1}{3}$

   Partial pressure of oxygen
   = mole fraction × Total Pressure

   $=\frac{1}{3}P$

6. The temperature at which real gases obey the ideal gas laws over a wide range of pressure is called Boyle temperature

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